The recipe of AI data centers: an engineering-based description by generation

28 September 2026. Working note, P&P project. Synthesis of `notes/recipe_research/` (four topic reports plus three helper reports, about 1,070 sourced observations). Companion to `recipe_plan.md`.

1 The recipe of the AI sectors

This note describes what it takes to build and run an AI data center. It gives a cost recipe for five generations of facilities, from pre-2015 enterprise halls to 2027+ designs. For each, it covers the facility, the IT equipment and the running costs, down to 6-digit NAICS suppliers, so the same recipe can feed the 37-sector model and the 2017 detailed benchmark tables. It prefers specialised sources (cost consultants, engineering white papers, national labs, standards bodies, company filings, measurement studies) and says where independent sources agree.

Seven results change the recipe. They are implemented in `ai_recipe.py`, in the project's data-construction folder (baseline A3). The 2026 Bank of Italy rack recipe is kept as `ai_recipe_cnt2026.py`:

1. The facility investment column needs installation labour. It becomes roughly Construction 49, Electrical 26, Machinery 19, professional services 6.5, not 33/40/27.
2. Facility running costs other than electricity are mostly repairs, taxes, software, security and insurance. Own labour is 10–16 percent of them, not 50.
3. Water is negligible in new designs.
4. Facility depreciation is 0.078 (0.085–0.09 if plant depreciates at 0.10), not 0.075.
5. IT capital must include networking and storage, 20–25 percent on top of the rack.
6. IT depreciation of about 0.26 is supported. It is driven by obsolescence, not failure.
7. The wholesale margin on IT purchases is about 6 percent, not 22.

Newer facilities trade water and labour for capital. They cost more per MW, use almost no water, and need a tenth of the staff of legacy halls.

1.1 Scope, sources and generations

Objects. Two sectors share out the cost of AI services:

- **AI_facilities** (f) owns the shell and the electrical and mechanical plant, buys electricity and runs the building, and sells powered space.
- **AI_compute** (z) owns the servers, networking and storage, buys space, and sells AI services.

A recipe is each sector's purchases by supplying industry, its labour, its capital by asset and service life, and its investment column (which industries produce its capital goods). It is expressed per MW of IT load; Section 1.6 converts it to per $1.

Source hierarchy. In order of preference:

1. Cost consultants, engineering and vendor white papers, national labs and standards bodies, specialist analysts, and measurement studies.
2. Company filings.
3. Economics papers, only when they are primary.

Each observation in the CSV files records whether it is primary or a re-quote, and traces re-quotes to their origin. Anything not checked at source is marked UNVERIFIED there.

Generations. Table 1 gives the five generations, with specifications corrected against the sources.

Table 1: Data-center generations

Code	Generation	kW per rack	Cooling	IT
A0	Legacy enterprise and retail colocation, pre-2015	4–8	Chillers, CRAH units, cooling towers	2-socket CPU servers
A1	Hyperscale cloud, air-cooled, 2015–22	8–15	Outside air, evaporative or adiabatic	CPU servers, storage servers
A2	AI, air-cooled or hybrid, 2023–24	≈40	Rear-door heat exchangers	HGX H100 8-GPU servers
A3	AI, liquid-cooled, 2025–26	120–145	Direct-to-chip, closed loop; dry coolers or chillers	GB200 / GB300 NVL72
A4	Next generation, 2027+	330 (Rubin NVL72) to 600 (Kyber)	Liquid; 800 V DC distribution	Vera Rubin

1.2 What changes relative to the 2026 recipe

Table 2: Current recipe against the evidence

Item	2026 rack recipe	Evidence	Main sources
IT vs facility share of capex	67/33 (one rack sheet)	68–72/28–32. Confirmed; rises to ≈72 once networking is included	Van Nieuwerburgh; Epoch; Fed FEDS 2025-109; Microsoft filings; SemiAnalysis
Facility cost per MW (A3)	Implicit in the rack sheet	$12m hard (10.5–14), $15m all-in (12–18). US mean rose from $8.9m (2021) to $11.1m (2025)	Turner & Townsend 2019–25; Digital Realty; KPMG 2026; JLL

Item	Now	Evidence	Main sources
Facility investment column	Construction 33, Electrical 40, Machinery 27	Construction 49, Electrical 26, Machinery 19, Prof. services 6.5 (A3). The rack sheet omits installation labour and fees	T&T element shares; KPMG owner-furnished split; Schneider WP61, WP218
Facility δ	0.075	0.078 (plant 0.088, shell 0.0247). Plant lives 18–20 years; shell about 15–20% of cost	ASHRAE; BEA; IRS; filings
Facility running costs other than electricity	50/25/25 labour, Admin, Prof. services; unsourced	About 0.6 per $1 of electricity. Repairs 0.19, taxes 0.13, own labour 0.10, software/IT 0.08, Admin 0.04, insurance 0.02	KPMG; Epoch; Digital Realty; Virginia JLARC
Water	Ignored	Near zero in A3 with dry coolers (0.05 L/kWh). About 3% of electricity cost in legacy halls. Price about $6 per 1,000 gallons	LBNL 2024; The Green Grid; Mytton 2021; operator reports
Electricity	$105k per rack-year	$640k per MW-IT-year for A3 (≈$85k per 132 kW rack): PUE 1.15, utilisation 0.75, $85/MWh. About 25% of facility cost	Operator PUEs; EPRI; LBNL; EIA
IT capital	Rack only	Rack ×1.25: scale-out network 14%, storage 4%, integration 3% of IT capex	SemiAnalysis; Cottier et al.; Epoch
IT δ	0.25	0.26 (0.20–0.32): GPU servers 0.27, networking 0.20, storage 0.30. Failure adds <1 point	31 SEC filings; rent indices; resale; Epoch; Meta and NCSA reliability
IT investment margins	Wholesale 22%	Wholesale 6%, transport 1% (direct purchase from manufacturers)	BEA PEQ; distributor margins; operator practice
Compute's other purchases	None	0.06–0.08 per $ of GPU-cloud revenue: software 0.02, telecom 0.01, maintenance 0.015, services 0.025	CoreWeave, Nebius, IREN filings; Census SAS 518

1.3 Facility capital

1.3.1 Cost per MW and element shares

Outside the hard cost, as shares of the A3 all-in cost:

- land 3–5 percent (Cushman & Wakefield: $584k/MW);

Table 3: Facility capex per MW of IT load, US, $m

	A0	A1 (2019/22)	A2	A3	A4 (projection)
Hard cost (T&T scope)	9 (6–13)	8.5 / 9.8	10.6 (9–13)	12 (10.5–14)	13–16
All-in (adds land, site, substation, fees, contingency)	11 (8–30)	10 / 11.5	12 (10–14)	15 (12–18)	16–22
Share of hard cost					
Electrical, incl. equipment	45%	54%	52%	48%	45–55%
Mechanical, incl. equipment	25%	22%	25%	33%	30–35%
Shell, core, architectural	20%	14%	12%	9%	7–9%
General contractor preliminaries and fees	10%	10%	10%	10%	8–10%

- site works 3–6 percent;
- the customer substation 1–3 percent;
- professional fees for design, engineering, project management and commissioning 5–8 percent;
- contingency 3–8 percent.

The liquid-cooling premium is 7–10 percent (T&T; JLL 10). Schneider finds about zero at equal rack density. Build time is about 18–20 months (JLL; Fed/Dodge).

1.3.2 The investment column at detailed codes

Equipment versus installation. The rack-based column puts all electrical and cooling spending on manufactured equipment. The engineering sources split it:

- Electrical: 50–60 percent equipment, 25–35 installation, 15–20 design (Schneider WP61). Materials are 71 percent of electrical cost in WP218. KPMG's split of owner-furnished equipment agrees.
- Mechanical: about 65 percent equipment.

Table 4 builds the A3 column per $15m of all-in cost.

Air-cooled halls. For A1, T&T's air-cooled shares with the same equipment ratios give Construction 53, Electrical 29, Machinery 12, Prof. services 6. With a 70 percent equipment share, A3 becomes 43/30/20/6.5.

Electrical equipment by maker. Shares of electrical equipment, A3 (A1 in brackets):

- generators, 335312 (engines 333618): 29 (31);
- UPS, PDUs and miscellaneous, 335999: 22 (21);
- switchgear, switchboards and busway, 335313: 22 (22);
- transformers, 335311: 9 (8);
- cable and wire, 335929: 7 (7);
- batteries, 335911: 6 (5);
- controls and other, 335314 and 334513: 5.

These replace the unsourced 30/20/15/15/10/5/5 in `upstream_concentration.py`, which gave too much to switchgear, transformers and batteries.

How electrical capex compares by generation. Installed electrical capex is about $5–6m per MW in every generation (A3 $6.0m, range 4.5–8). The bottom-up build, from Schneider unit costs indexed by the producer price index (PPI), comes in 15–25 percent below the consultants. PPIs miss the 2023–26 backlog premiums. Transformer prices rose 76 percent from 2019 to 2025 (PPI; Wood Mackenzie and NIAC agree), and lead times are 120–150 weeks.

Table 4: AI_facilities investment column, A3, per MW of IT load

Element	$m	Producer (NAICS)
Shell, core, architectural	1.08	Construction (BEA 2332A0)
Contractor preliminaries and fees	1.20	Construction
Electrical installation (40%)	2.30	Construction (238210 electrical contractors)
Mechanical installation (35%)	1.39	Construction (238220)
Site works and civil	0.60	Construction
Customer substation	0.30	Half Construction, half 335311/335313
Electrical equipment (60%)	3.46	See the split below
Mechanical equipment (65%)	2.57	333415 HVAC, chillers, CRAHs, CDUs; 333914 pumps; 333413 fans; some 332 pipe and manifolds
Design, engineering, commissioning	0.90	541330, 5413
Contingency	0.60	Pro rata
Land	0.60	Not produced; excluded
Produced total	**14.4**	**Construction 49%, Electrical 26%, Machinery 19%, Prof. services 6.5%**

Imports of electrical equipment, 2024–25 (share of value imported):

Equipment	Best (range)	Note
Transformers	40% (30–55)	Large power transformers about 75–80%; distribution transformers 25–50%
Switchgear	25% (15–45)	Medium-voltage switchgear 9–15%
Generator sets	30% (20–50)	BEA's 64% looks too high for large diesel sets
UPS	50%	
Lithium-ion batteries	50%	

How the national accounts record it. This matters for decision R4:

- Census C30 and BEA treat integral electrical and mechanical plant as structures.
- BEA added a data-center line to NIPA Table 5.4 in September 2025.
- The UK statistics office (ONS, 2026) books racks, PDUs and stand-alone backup generators as machinery.

The model's Construction sector has the 2017 average recipe, which buys almost no switchgear or transformers. Routing equipment directly to its producers therefore preserves the supply chain.

1.4 Facility operations

1.4.1 Electricity

Energy per MW of IT is roughly flat after the legacy generation, because rising utilisation offsets falling PUE. Hyperscaler fleet PUEs agree with each other: Google 1.09, Meta 1.08, AWS 1.14–1.15, Microsoft 1.16–1.17. Industry surveys are higher (Uptime 1.36) because they include colocation and enterprise halls. No measured PUE exists yet for a GB200-class site.

Table 5: Electricity per MW of IT load per year

	A0	A1	A2	A3	A4
Operating PUE	1.70	1.15	1.25	1.15	1.10
Utilisation (average draw / provisioned)	0.50	0.65	0.70	0.75	0.80
GWh per MW-IT-year	7.4	6.5	7.7	7.6	7.7
Price, $/MWh	110	75	80	85	85
Cost, $k per MW-IT-year	**819**	**491**	**613**	**642**	**655**

On-site generation is about 90 GW announced but only 2 GW operating. Treat it as a scenario and as utility-type capital (turbines and engines, 333611 and 333618; gas, 211000 and 486000), not as part of AI_facilities.

1.4.2 Water: old versus new designs

Table 6: Water per MW of IT load

	A0	A1	A2	A3	A4
On-site use, L per kWh of IT energy	2.0	0.30	0.50	0.05	0.02
Water and sewer cost, $k per MW-year	26	4	7	0.6	0.2
Water cost / electricity cost	3.4%	0.7%	1.1%	0.1%	≈0

New designs cut water in two steps:

- Outside-air hyperscale cooling cut on-site water 5–10 times at almost no energy cost.
- Warm-water liquid cooling with dry coolers takes it to about zero with no energy penalty: PUE 1.10, against 1.14 with cooling towers.

The saving comes from the heat-rejection choice, not from liquid cooling as such. An A3 hall on cooling towers still uses about 2 L/kWh.

Legacy halls used towers because going waterless cost about 0.3 points of PUE. That is roughly $138k of extra electricity to save $38k of water per MW-year.

Counting the water used to generate the electricity (about 4.4 L/kWh), total water per unit of IT falls only 35–40 percent from A0 to A3.

The "industry average 1.8 L/kWh" often quoted is a 2016 LBNL assumption per kWh of total site energy, not a measurement.

1.4.3 Labour, maintenance and other running costs

Staffing agrees across sources at about 1.3 FTE per MW for a 30–40 MW hall (Virginia JLARC, KPMG, Uptime 2007). At gigawatt scale the sources diverge, from Meta at 0.2 to operator announcements at 0.9–1.5. BLS employment in NAICS 518210 is not a measure of facility staff. Most facility staff work for the parent company or for facilities-support and security firms.

Table 7: Facility running costs other than electricity, $k per MW of IT load per year

	A0	A1	A2	A3	A4	Supplier
Facility staff, FTE per MW	5	1.3 per hall; 0.6 per campus	0.7	0.5	0.3	Half own staff; half 5612/5616/5617
Own payroll	400	133	90	60	40	Labour
Plant maintenance	150	70	100	120	150	230301 repairs; some 811300
Security	150	43	30	20	15	5616
Cleaning	40	13	7	5	3	5617
Insurance	10	10	12	15	20	524
Property tax, after abatements	80	50	70	80	95	Tax on production
Software, IT and network operations	100	60	50	50	55	517, 518, 5415
Other (generator fuel and testing, consumables)	40	25	25	30	30	324110, misc.
Total	**∼1,000**	**∼410**	**∼390**	**∼380**	**∼410**	

1.4.4 Service lives

- Shell: about 40 years (BEA offices: 36 years, $\delta = 0.0247$).
- Electrical switchgear, transformers and busway: about 30 years.
- UPS electronics: about 10 years. Lithium-ion batteries: 12. VRLA batteries: 4, which is a replacement flow in legacy halls.
- Mechanical plant: 18–20 years.
- Blended facility δ: 0.078 with plant at 0.088 (BEA's 1.65/L rule with a 19-year life); 0.085–0.09 with plant at 0.10.

1.5 IT capital

Table 8: IT capital per generation

	A0/A1	A2 H100	A3 GB200	A3 GB300	A4 Rubin
Rack or server price	$7.5–10k per server	$190–270k per server	$3.1m	$4.5m	∼$7m
All-in per GPU	–	$35k	$54k	$76k	∼$115k
$m per MW of IT	**15** (10–22)	**24** (21–28)	**27** (26–30)	**37** (31–42)	∼**50** (35–65)
Network share of IT capex	15%	15%	14%	13%	12%
Storage share	15%	4%	4%	4%	5–15%
Integration share	3%	3%	3%	3%	2–4%
NVIDIA gross profit / IT capex	–	∼55%	55–60%	55–60%	55–60%
Geometric δ	0.30	0.27	0.27	0.27	0.27–0.33
Import share	0.85–0.95 (servers); networking 0.80–0.95; storage 0.60–0.80				

Confidence is highest for H100 and GB200, which have several primary analyst sources, vendor specifications and filings. It is medium for GB300 and low for A0/A1 per MW and for A4.

Cost per rack rose about 15 times from CPU racks to GB200, but cost per MW only about 2 times, because rack power rose about 12 times.

Composition, A3. Shares of cluster IT capex:

- GPU modules including HBM: 55;

- the rest of the compute tray: 14;
- NVLink switches and rack infrastructure: 10;
- scale-out network: 14;
- storage: 4;
- integration: 3.

NVIDIA content is over 92 percent of the rack, at a 71–75 percent gross margin. With a closed economy, that margin becomes value added of Computers. This is why the import treatment matters.

Investment column of AI_compute, at purchasers' value. Central values (range):

- computers (334111, 334112): 0.77;
- communications equipment (334210; networking): 0.13;
- integration, computer systems design (541512): 0.03 (0.02–0.05);
- wholesale: 0.06 (0.03–0.10);
- transport: 0.01;
- retail: 0.

This replaces the average-buyer margins in the BEA 2017 bridge, where wholesale is 0.19. US server imports reached $164bn in 2025, 91 percent from Mexico and Taiwan. 58 percent of server exports are re-exports: GPUs and boards go out and come back as servers.

Depreciation. Five independent kinds of evidence give $\delta \approx 0.20$–0.30 for GPU servers:

- **Accounting lives** of 5–6 years in 31 SEC filings. Amazon cut part of its fleet to 5 years in 2025, citing AI. Neoclouds use 4–6 years.
- **Rents across chip generations:** about 0.27 log points per year of vintage.
- **Resale prices by age:** 0.18–0.28.
- **Price-performance:** 0.28–0.40 log per year (Epoch).
- **Physical failure is small:** replacements of 0.4–0.8 percent a year. The widely quoted 9 percent counts job interruptions, not dead GPUs.

The 2023–25 fall in H100 rents (0.35–0.48 a year) includes the unwinding of a scarcity premium and should not be read as δ.

Compute's operating inputs other than power and space. Per $1 of GPU-cloud revenue:

- software: 0.02;
- telecom and transit: 0.01;
- maintenance and spares: 0.015;
- purchased sales, marketing and administrative services: 0.025;
- labour, excluding the labs: 0.06–0.12;
- property tax: about 0.01, a production tax.

Sources: CoreWeave, Nebius and IREN filings. The Bank of Italy's network, maintenance and "other" lines are 2–4 percent of CoreWeave's realised revenue per rack, which is consistent. Its list rental prices are 4–6 times realised revenue.

1.6 Per $1 of AI services

The recipe is scaled from per MW of IT load to $1 of AI services in three steps:

1. The facility's output (powered space) is its capital at user cost $r + \delta_f$, plus electricity, other purchases, labour and property tax.

2. Compute costs its IT capital at user cost $r + \delta_z$, plus space, plus operating inputs (purchases 6.5, staff 9 and property tax 1 percent of its cost).
3. Korinek and McKelvey's closure turns compute into AI services: services revenue is 1.5 times inference compute, the labs' labour is 0.1 times inference, and the margin pays for training compute. Compute is therefore 0.933 of each $1.

Table 9 reports the result by generation, with the 2026 rack recipe for comparison.

Table 9: Recipe per $1 of AI services, by generation

	A0	A1	A2	A3	A3 GB300	A4	2026 rack
z: labour (labs + staff running the computers)	0.151	0.151	0.151	0.151	0.151	0.151	0.067
z: IT capital services	0.487	0.537	0.588	0.589	0.632	0.650	0.669
z: space from f	0.297	0.247	0.196	0.195	0.152	0.134	0.264
z: other purchases	0.065	0.065	0.065	0.065	0.065	0.065	0
f: electricity	0.080	0.053	0.049	0.046	0.036	0.027	0.080
f: other purchases	0.050	0.024	0.018	0.017	0.013	0.011	0.032
f: labour	0.039	0.014	0.007	0.004	0.003	0.002	0.032
f: capital services	0.128	0.155	0.122	0.128	0.100	0.093	0.120
IT capital per $1 a year	1.49	1.65	1.94	1.94	2.08	2.13	2.29
Facility capital per $1 a year	1.12	1.31	1.05	1.07	0.83	0.76	1.03
δ_{IT} / δ_f	.285/.072	.285/.076	.261/.074	.261/.078	.262/.078	.264/.081	.25/.075
Electricity share of f's cost	0.27	0.21	0.25	0.23	0.23	0.21	0.30
AI services per MW-IT-year, $m	10.3	9.3	12.5	14.1	18.1	23.8	–
f investment: Constr./Elec./Mach./Prof.	55/25/13/7	52/29/11/7	51/28/13/7	49/26/17/7	49/26/17/7	48/27/17/7	33/40/27/0
z investment: Computers/integration/wholesale			90 / 3 / 6 (transport 1)				78/0/22

Newer generations spend a larger share of each dollar on IT capital and a smaller share on space. Electricity falls from 8 cents per dollar (legacy) to under 3 (next generation), because chips get more expensive per watt faster than power gets cheaper.

Two decisions are kept unchanged for the authors (`recipe_plan.md`, R1–R2): the labs' labour (Korinek and McKelvey's 0.1 of inference, which the evidence puts about 10 times too low) and the treatment of training as a current cost.

One cross-check is open. At user cost on new capital, compute costs about $13m per MW-year. Epoch's total cost of ownership and CoreWeave's revenue per MW are about $8.5m.

1.7 Cross-checks

Table 10: Where independent sources agree and disagree

Number	Sources	Agree?	Why they differ / verdict
IT share of campus capex	Van Nieuwerburgh 68; Epoch 69; Fed 70 (IT-inclusive); Microsoft PP&E 57 servers; CNT rack + network 72	Yes	68–72
Facility hard cost per MW	T&T; JLL; KPMG; Schneider	Yes, once scope and year match	$12m (A3)
Facility all-in cost per MW, A3	Landlords 8–13; DLR 14.1; C&W 17.6	Spread	Scope: brownfield vs land, soft costs and contingency

Number	Sources	Agree?	Verdict
Shell share	9–35% across sources	No	Some include site work; use 9–14% plus 3–6% site
Equipment vs installation	Schneider WP61 (only primary); KPMG; Eaton content	Single source	Gap narrowed, not closed
Electrical capex per MW	Bottom-up PPI build; T&T × share; C&W	Partly	PPIs miss backlog premiums; lean to the consultants
Eaton AI content	\$2.4m/MW vs bottom-up \$1.3m	No	Possibly IT-side power; unresolved
Transformer price inflation	PPI +76%; Wood Mackenzie +77–95%; NIAC +80%	Yes	
Transformer import share	Own calculation 26.5% (2017), ≥38.5% (2024); BEA 29%; BIS 82% of large units	Yes by size class	
Generator import share	Own calculation ≤44%; BEA 64%	No	BEA too high for large sets
Hyperscale PUE	Google, Meta, AWS, Microsoft 1.08–1.17; LBNL 1.22; Uptime 1.36	Operators yes	Surveys include colocation and enterprise
Hyperscale water (L/kWh)	LBNL 0.32; Microsoft 0.27–0.30; Meta 0.20; Virginia utilities 0.2–0.3	Yes	Withdrawal vs consumption explains a 1.3× gap
Staff per MW, 30 MW hall	JLARC; KPMG; Uptime	Yes	1.3
Staff per MW, 1 GW campus	Meta 0.2; Crusoe 0.47; Oracle and Microsoft announcements 0.9–1.5	No	0.5 (0.3–1.0)
Facility running costs other than electricity	KPMG; ITK; Epoch; Digital Realty 0.33–0.41m; CNT 0.64m	Yes except CNT	CNT includes network (belongs to z) and an unsourced line
Electricity share of facility cost	Own A3 25%; engineering notes 25–30%; Digital Realty 23% of revenue	Yes	
GB200 rack price	CNT \$3.0m; SemiAnalysis \$3.1m	Yes	Independent
Non-rack share of IT	SemiAnalysis (2 articles); Cottier; Epoch	Yes	Van Nieuwerburgh's 12.5% is low and unsourced
GB300 price	Van Nieuwerburgh \$4.0m; purchase orders \$5.0m; press \$6–6.5m	No	±25%
IT δ	SEC lives; rents; resale; Epoch	Yes	0.26 (0.20–0.32)
Wholesale margin on servers	BEA PEQ 19%; distributors 7–11%; direct purchase	No	6% for hyperscalers

Mis-quotes found and corrected.

- "1.8 L/kWh industry average": a 2016 LBNL assumption, not a measurement.
- "9 percent annual GPU failure rate": a press calculation on Meta's interruption table. It counts interruptions, and the paper itself has a typo in the table.
- "GPUs last 1–3 years": an anonymous social-media post.
- Van Nieuwerburgh applies a per-IT-MW cost to total facility power.
- Epoch's facility cost re-quotes Turner & Townsend.
- IEA's "84 percent training load factor" is not in the LBNL report it cites.

1.8 Imports of AI capital: IT equipment, facility gear and the grid

The current treatment. The baseline economy is closed, following the vom Lehn–Winberry convention. The paper's imports alternative opens one flow only:

- Imports supply 95 percent of what AI_compute buys from Computers, the BEA 2017 import share of computers.
- A linear technology turns an export basket with the composition of 2017 consumption into imported equipment at a fixed rate.
- Trade balances every year.

Facility gear (electrical and mechanical equipment) and the grid equipment that Utilities buys stay domestic. The paper's conclusion lists imports of data-center electrical equipment as a next step. This section sets out how to add them. Nothing here is implemented yet.

How big the flows are. In the steady state of the A3 recipe, per $1 of AI services:

- AI_facilities invests $\delta_f K_f \approx 0.080$ a year: 0.021 in electrical equipment and 0.013 in machinery.
- AI_compute invests $\delta_z K_z \approx 0.50$ a year, of which 0.45 is Computers (servers, networking and storage).

At the import shares below, imported facility gear is about 0.010 per $1 of AI services, about 1 percent of AI revenue. Imported IT is about 0.40. Gear imports therefore barely move GDP, consumption or AI investment.

They matter for industries. Electrical equipment is one of the paper's headline winners, and in the IT-imports alternative it is the industry that takes the place of Computers. If about 40 percent of the AI-driven demand for electrical gear goes abroad, that gain shrinks accordingly.

How to implement it in the model. Generalise the single import weight to one weight m_{ij} for each pair of supplying sector i and investing sector j. The price of j's investment bundle becomes

$$\tilde{P}_j^k = \Big(\sum_i (1 - m_{ij})\, G_{ij}^I\, P_i^{1-\sigma_I} + \Big(\sum_i m_{ij} G_{ij}^I \Big) P_{imp}^{1-\sigma_I} \Big)^{1/(1-\sigma_I)},$$

where P_{imp} is the price of the export basket that pays for imports. The rest follows directly:

- Imports by sector j are $I_j^{imp} = \big(\sum_i m_{ij} G_{ij}^I\big)(P_{imp}/\tilde{P}_j^k)^{-\sigma_I} I_j$.
- Exports are the consumption-weighted basket that pays for $\sum_j I_j^{imp}$.
- The resource constraint and GDP add up over j.

In the code, the scalar import weight becomes a matrix, filled from import shares stored by NAICS code in `ai_recipe.py` and aggregated with the recipe. Two choices remain:

1. **Vintage of the shares.** BEA 2017 detail is consistent with the vintage of the current IT exercise. The 2024–25 trade data match a 2025–26 recipe. The weighted electrical share is about 0.50 on BEA 2017 and 0.40 on 2024–25; the main difference is generator sets. Using 2024–25 for both IT and gear moves the IT share only from 0.95 to about 0.90.

Table 11: Import shares of AI capital (share of value imported)

Item	NAICS	BEA 2017	2024–25 (best, range)	Note
AI_compute: IT equipment				
Servers	334111	0.95	0.90 (0.85–0.95)	Mexico and Taiwan are 91% of 2025 server imports
Networking	334210	–	0.875 (0.80–0.95)	Vietnam, Thailand
Storage	334112	–	0.70 (0.60–0.80)	Thailand
AI_facilities: electrical equipment (share of it in brackets)				
Generator sets (29%)	335312	0.64	0.30 (0.20–0.50)	BEA too high for large diesel sets
UPS, PDUs (22%)	335999	0.55	0.50 (0.35–0.60)	Proxy: static converters
Switchgear, busway (22%)	335313	0.26	0.25 (0.15–0.45)	Medium voltage 9–15%
Transformers (9%)	335311	0.29	0.40 (0.30–0.55)	≥38.5% in 2024; large units 75–82%
Cable and wire (7%)	335929	0.91	not re-checked	BEA 2017 value
Batteries (6%)	335911	≈0.36	0.50 (0.40–0.70)	Li-ion; VRLA ≈0.35
Controls (5%)	335314	–	≈0.45	Broad switchgear and controls, 43–50%
Weighted electrical		**≈0.50**	**≈0.40**	
AI_facilities: mechanical equipment				
HVAC, chillers, CDUs	333415	0.14	≈0.15	Pumps and fans not studied
Utilities: grid equipment				
Large power transformers	335311	–	0.75–0.82	BIS 2020; Wood Mackenzie
Distribution transformers	335311	–	0.20–0.50	BIS 18–21% (2019) vs Wood Mackenzie 50%
Medium- and high-voltage switchgear	335313	0.26	0.15	

2. **Utilities.** The grid bottleneck runs through the equipment Utilities buys to expand transmission and distribution: transformers and switchgear, 15 percent of its 2017 investment column. Importing them means opening an existing sector's investment column, which departs from the closed-economy convention for the 37 sectors. There are two ways to do it:
   - *Average shares.* Utilities imports its electrical equipment at all dates, including 2017. Then the 2017 steady state has trade, and the export basket must be calibrated to it.
   - *Marginal shares.* Only purchases above the 2017 level are imported. This keeps the 2017 calibration unchanged but is less standard.

   The first is cleaner and makes the grid, and its import dependence, part of the exercise. The authors plan to include Utilities if the extension proceeds.

What it would add to the paper.

- A version of the imports alternative with IT, facility gear and grid equipment.
- How much of the Electrical sector's gain leaks abroad.
- Whether the ranking of winners survives.
- A natural tariff scenario: a tax on imported AI capital. See the Quintana et al. (2026) paper on tariffs as taxes on capital in `PandP/`.

Run time is about 10 minutes per scenario.

1.9 Using the recipe with the 2017 detailed tables

The recipe is specified at 6-digit NAICS wherever the sources allow. The proposal is to store it as a vector over the 2017 detailed commodities:

- one vector per generation and per sector (running costs and investment column);
- with the source, year and range attached to each cell;
- and let the loader aggregate it to the 37 sectors.

The same data then serve the paper as it stands, a detailed or split version of the model (the `Construction_AI`-type splits already in `model.py`), and import shares by commodity, which differ a lot within Electrical.

Two caveats:

- Several NAICS codes map to aggregated BEA detail codes, e.g. pumps and batteries. The NAICS-to-BEA mapping is still to be done.
- The 2017 tables have no data-center industry. Hosting (518200) is mostly payment processing, and data-center buildings sit in 2332A0. The detailed tables supply the suppliers, not a data-center column to copy.

1.10 Open gaps

- Equipment versus installation rests on one primary source (Schneider WP61).
- No measured PUE or water use for GB200-class sites. No cost data on liquid-cooling maintenance.
- The split of mechanical equipment among HVAC, pumps and fans is not quantified.
- Insurance rates; the size of owner-furnished equipment beyond KPMG; grid-upgrade cost per kW.
- Prices for GB300 and Rubin rest on press reports and one blog. SemiAnalysis item-level tables are paywalled.
- The share of hyperscaler servers bought directly from manufacturers.
- The labs' cost structure (decisions R1–R2).

1.11 Main sources

- **Cost consultants:** Turner & Townsend, Data Centre Construction Cost Index (2019–25); KPMG (2026), benchmarking capex and opex; JLL 2026 outlooks; Cushman & Wakefield (partly verified).
- **Engineering and vendors:** Schneider Electric white papers WP61, WP75, WP218, WP229 and WP282; Uptime Institute (Koomey et al. 2008; annual surveys); Eaton; NVIDIA DGX and NVL72 specifications and its 800 V DC paper.
- **National labs, agencies and standards:** LBNL (Shehabi et al. 2024); EPRI; EIA; The Green Grid WP35; ASHRAE; BIS and DOE on transformers; Mytton (2021); Li et al. (2023).
- **Official and filings:** Virginia JLARC (2024); BEA depreciation rates and NIPA 5.4; ONS (2026); UN Comtrade and Census trade data; SEC filings of Amazon, Microsoft, Alphabet, Meta, Oracle, CoreWeave, Nebius, IREN, WhiteFiber, Nscale, Digital Realty and Equinix.
- **Analysts and measurement:** SemiAnalysis (GB200 architecture, AI cloud total cost of ownership (TCO), GPU index); Epoch AI (hardware trends, 1 GW TCO); Silicon Data and Ornn rent indices; Meta, NCSA, BigScience and ByteDance reliability studies; Cottier et al. (2025).

Files:

- Reports and observations: `notes/recipe_research/*.md` and `*_obs.csv`.
- PDFs: `References/engineering/`, in four subfolders (facility capex, power and energy, operations, IT equipment).